

From Regime Boundaries to Chemical Gauge Group: The Inevitable Emergence of the Periodic Law and the Cross-Disciplinary Unification of the Factor Hierarchy Law

——Derivation of the Chemical Gauge Group Theorem, the Chemical Counterpart of the Higgs Mechanism, and the Establishment of the Five-Layer Axiomatic Structure (III)

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Abstract

Why are there exactly four types of atomic orbitals—s, p, d, and f? Why does the periodic table have an end? What is the relationship between the Factor Hierarchy Law, quantum mechanics, and gauge field theory? Building on the empirical discoveries of the two preceding papers, this paper completes three core theoretical derivations that elevate the periodic law from an empirical regularity to an inevitable deduction from first principles. First derivation: the Chemical Gauge Group Theorem. The meta-regime factor valence_electrons possesses $n=8$ mutually exclusive regime states (outermost electron numbers = 1, 2, ..., 8), and its moderating effects cover all executive factors with irreducible interaction effects. Under the symmetry constraint of the three-dimensional rotation group $SO(3)$, the unique group-theoretic correspondence of the eight regime states is the $l=0, 1, 2, 3$ irreducible representations of $SO(3)$ —namely, the s, p, d, and f orbitals. The sum of the dimensions of these irreducible representations is $1+3+5+7=16$, exactly corresponding to eight spin-degenerate regime states. No regime state with $l \geq 4$ exists, because the $l=4$ representation is 9-dimensional, corresponding to 18 spin-orbitals, far exceeding the capacity of eight regime states, and the required nuclear charge number would exceed the nuclear stability limit. Second derivation: the division of labor and subsumption between the Factor Hierarchy Law and quantum mechanics. The Factor Hierarchy Law provides the regime-state structure (the eight mutually exclusive regime states of valence_electrons), and quantum mechanics provides the symmetry constraint (the underlying symmetry of electron configuration is $SO(3)$). Together, they derive the inevitability of s, p, d, and f orbitals. This is not “dependence” on quantum mechanics but rather the demotion of quantum mechanics from a “fundamental theory” to “a provider of symmetry constraints as physical input”—completely isomorphic in logical structure to the “demotion of gauge field theory” in Tang (2026q). Third derivation: the end of the periodic table. The topological constraints of transition bandwidths determine the upper limit of the number of regime centers. As the orbital angular momentum l increases, the spacing between regime centers expands exponentially. The entrance to the g-block ($l=4$) at $Z \approx 121$ is the last detectable regime center. The nuclear charge number required for the entrance to the h-block ($l=5$) exceeds the nuclear stability limit—the fission barrier height drops below 1 MeV, and any heavier nuclide would fission at the moment of formation.

The end of the periodic table is not an arbitrary human stipulation but the inevitable joint consequence of topological constraints from regime centers and the physical limits of nuclear stability. Integrating these three derivations, this paper establishes the five-layer axiomatic structure of the regime-switching theory of the periodic law—Law Layer (three axioms) → Norm Layer (six functional dimensions) → Algorithm Layer (three-step MRSD logic) → Theorem and Proposition Layer (four quantitative laws + the Chemical Gauge Group Theorem) → Theory Layer (nine cross-disciplinary corollaries). This structure is completely isomorphic to the five-layer structure established by Tang (2026q) in particle physics. The Factor Hierarchy Law emerged from financial data, was verified in stellar atmospheres, obtained its mathematical foundation in statistical physics, derived gauge group structures in particle physics, and completed the demotion of fundamental laws in chemistry—it is not the law of a single discipline but the universal logic of the functional stratification of variables in any complex system based on objective data.

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1 Introduction

1.1 Deep Unanswered Questions in Chemistry

The periodic table is one of the most familiar scientific tools of humanity. Since Mendeleev published the first periodic table in 1869, more than 150 years of research have refined its empirical regularities to the extreme. However, two fundamental questions have never been answered.

The first question: why are there exactly four types of atomic orbitals—s, p, d, and f? Quantum mechanics solved the spherical harmonics from the Schrödinger equation, and the orbital angular momentum quantum number l of spherical harmonics can take any non-negative integer— $l=0, 1, 2, 3, 4, \dots$. Mathematically, there is no upper bound on l . However, in physical reality, we only observe four types of orbitals with $l=0, 1, 2, 3$ —s, p, d, and f. Why do $l=4$ (g orbitals) and higher orbitals not exist in nature? The traditional answer is “because the 118 known elements do not require g orbitals to fill electrons.” But this is a circular argument—it does not explain why the number of elements happens to terminate precisely at the range that does not require g orbitals.

The second question: why does the periodic table have an end? The traditional answer is “because atomic nuclei are unstable in the superheavy element region.” But this merely shifts the problem to nuclear physics—why does nuclear stability happen to terminate at a position corresponding to the end of the

periodic table? Is there a deeper principle that simultaneously constrains both electron configuration and nuclear stability?

1.2 Review of the Empirical Foundations of the Two Preceding Papers

Tang (2026s) deployed the Factor Hierarchy Law and Testability Norms to conduct a presupposition-free regime audit of the 118 elements, yielding seven core laws. The most critical for the theoretical derivations in this paper are: (1) period and valence_electrons are confirmed as two orthogonal regime factors in the periodic table—they do not directly explain chemical properties but reset the entire property-determination equation by moderating the exposure coefficients of executive factors; (2) at $Z=20$ (calcium, d-block entrance) and $Z=57$ (lanthanum, f-block entrance), the Chow F under inv transformation reaches 1.53×10^{23} and 3.37×10^{23} , respectively, with the f-block/d-block intensity ratio of 2.20 deviating only 2.2% from $(3/2)^2=2.25$, revealing for the first time the quantitative law $F(1) \propto I^2$ governing regime-switching intensity; (3) the regime transition bandwidths of four intrinsic properties are strictly unified at 3.0 periods.

Tang (2026t) built on these discoveries to construct the Regime Periodic Table (RPT), generated regime identity cards for the 118 elements, achieved 100% PRISMA coverage, completed a cross-sample convergence audit, constructed the Universal Regime Switching Operator (URSO), and provided purely deductive numerical predictions for elements 119 and 120: for 119, a density of 3.3–3.5 g/cm³ and an electronegativity rebound to 0.85–0.90; for 120, a density surge to 7.0–7.5 g/cm³ and a first ionization energy soaring to 5.8–6.1 eV, exhibiting pseudo-noble character.

1.3 The Theoretical Task of This Paper

This paper is the final installment of the three-part chemical series, covering the dimension of Why (why does it happen). Based on the full empirical discoveries of the two preceding papers, the core task of this paper is to complete three theoretical derivations that elevate the periodic law from an empirical regularity to an inevitable deduction from first principles.

First derivation: the Chemical Gauge Group Theorem—starting from the eight mutually exclusive regime states of valence_electrons, through the irreducible representations of SO(3), rigorously deriving the inevitability of s, p, d, and f orbitals.

Second derivation: the division of labor and subsumption between the Factor Hierarchy Law and quantum mechanics—explicitly identifying what the Factor Hierarchy Law provides, what quantum mechanics provides, and how the two together derive the inevitable emergence of the periodic law.

Third derivation: the end of the periodic table—starting from the topological constraints of transition bandwidths and the nuclear stability limit, demonstrating that the g-block is the last detectable regime center.

Integrating these three derivations, this paper establishes the five-layer axiomatic structure of the regime-switching theory of the periodic law, completes the cross-disciplinary isomorphic evidence matrix across five disciplines, and establishes the Factor Hierarchy Law as the ultimate “law of meta-laws.”

2 Review of the Axiom Foundations

This section briefly reviews the verification status of the three axioms in chemistry; for detailed arguments, see Tang (2026s).

Axiom 1 (Factor Dichotomy): period and valence_electrons are confirmed as two orthogonal regime factors in the periodic table. In exhaustive testing, period did not enter any optimal combination under $VIF < 5$; when VIF was relaxed to < 10 , it broke into the optimal combination of first_ionization_100kJmol—the canonical statistical signature of a regime factor. Electronegativity is confirmed as a dual-identity factor—it directly explains chemical properties while also being the only “super-probe” capable of marking the same regime-switching center across three regime axes simultaneously. The DAG causal network confirmed 13 causal paths, with the direct causal parent nodes of electronegativity being precisely valence_electrons, unfilled_d_orbitals, and unfilled_f_orbitals.

Axiom 2 (Regime Switching): At $Z=20$ (calcium, d-block entrance) and $Z=57$ (lanthanum, f-block entrance), the inv-transformed Chow F reaches 1.53×10^{23} and 3.37×10^{23} , respectively. The f-block entrance switching intensity is 2.20 times that of the d-block entrance, deviating only 2.2% from $(3/2)^2=2.25$. Ten structural break tools all detected significant breaks near the two centers, and all Bai-Perron sequential multi-break BIC improvements were positive (up to 380.2). The boundary concentration law of interaction effects further confirmed: four categories of boundary elements all show significant interaction effects ($p < 0.02$), whereas transition metals ($n=35$) and lanthanides ($n=15$), as the largest categories within regime interiors, are not significant—regime switching occurs at boundaries, not in interiors.

Axiom 3 (Interaction Effects): The complete interaction effect matrix yielded 14 significant results out of 27 tests ($p < 0.05$), with zero false positives in synthetic data negative controls (0/4). The density–atomic mass correlation plummeted from 0.7271 to 0.2809 across the period axis ($p=0.0055$), and the boiling point–ionization energy correlation reversed from 0.3010 to -0.5080 ($p=0.0008$). The function of regime factors is not embodied in direct explanatory power but in their interaction effects that moderate the exposure coefficients of executive factors.

The verification status of these three axioms in chemistry is completely consistent with their verification status in particle physics (Tang, 2026n-q) and statistical physics (Tang, 2026k-m). Together, they constitute the empirical foundation for the theoretical derivations in this paper.

3 The Hierarchical Structure of Meta-Regime Factors

3.1 Orbital Angular Momentum l Is the Meta-Regime Factor Hidden behind period and valence

The Calcium-Lanthanum Singularity Theorem discovered by Tang (2026s)— $F_f/F_d=2.20\approx(3/2)^2$ —is not merely a coincidence of two data points but points to a deeper physical reality: the intensity of regime switching is not determined by period or valence_electrons but by the orbital angular momentum l hidden behind these two observable dimensions.

In quantum mechanics, the orbital angular momentum l is the core quantum number describing the spatial distribution of electrons. $l=0$ corresponds to s orbitals (spherical), $l=1$ to p orbitals (dumbbell-shaped), $l=2$ to d orbitals (cloverleaf-shaped), and $l=3$ to f orbitals (complex multipolar shapes). period and valence_electrons are two orthogonal projections of l onto observable dimensions—period corresponds to the principal quantum number n (determining orbital size), and valence_electrons corresponds to the number of outermost electrons (determining orbital filling status). However, what truly controls the intensity of regime switching is l itself—because l directly determines the symmetry structure of the orbitals, and changes in symmetry structure are precisely the physical root of regime switching.

This interpretation rules out a natural alternative hypothesis: could the intensity of regime switching be determined by the electron capacity of the orbitals? According to the Pauli exclusion principle, the upper limit of the number of electrons per orbital is $2l+1$ —2 for s orbitals, 6 for p orbitals, 10 for d orbitals, and 14 for f orbitals. The electron capacity ratio for the f-block/d-block is $14/10=7/5=1.40$. However, the empirical $F_f/F_d=2.20$ is inconsistent with 1.40—ruling out the hypothesis that regime-switching intensity is determined by electron number capacity. That regime-switching intensity is proportional to l^2 rather than $2l+1$ implies that the intensity of singularity centers in variance-partition space is determined by the symmetry structure characterized by orbital angular momentum, not by the electron capacity of the orbitals.

3.2 The Empirical Basis of $F(l) \propto l^2$

The sole direct empirical evidence for $F(l) \propto l^2$ comes from the two data points of the d-block ($l=2$) and the f-block ($l=3$). The choice of this functional form requires further justification—why l^2 rather than some other function of l ?

In quantum mechanics, l^2 is the eigenvalue of the orbital angular momentum square operator: $L^2\psi = l(l+1)\hbar^2\psi$. In the large- l limit, $l(l+1)\approx l^2$. The spin-orbit coupling energy is proportional to l 's, with its energy scale proportional to l . The regime-switching intensity F measures the structural break intensity detected by the Chow test at regime boundaries—this is directly related to the “severity” of the change in orbital symmetry structure. The transition from $l=1$ to $l=2$ ($p\rightarrow d$) changes the parity and spatial form of the orbital, and the transition from $l=2$ to $l=3$ ($d\rightarrow f$) further increases the number of nodes of the orbital in space. This

change in symmetry structure is the physical root of regime switching, and l^2 precisely quantifies this change.

It must be candidly acknowledged that currently only two data points, $l=2$ and $l=3$, support $F(l) \propto l^2$. The universality of its functional form requires independent verification from g-block ($l=4$) experimental data—this is precisely the scientific significance of the prediction of the g-block Chow $F \approx 5.99 \times 10^{23}$ in Tang (2026t).

3.3 The Three-Level Hierarchy: $l \rightarrow$ period/valence $\rightarrow$ electronegativity

Based on the above analysis, this paper establishes the three-level hierarchical structure of meta-regime factors (Theorem 1, MA3-1).

Level 1: Meta-regime factor—orbital angular momentum l . l directly determines the symmetry structure of the orbitals and is the ultimate determinant of regime-switching intensity. l was not directly selected by exhaustive testing from the data—it emerged from the quantitative correspondence $F_f/F_d \approx (3/2)^2$. This quantitative correspondence is a canonical example of the Factor Hierarchy Law “discovering hidden structures behind known phenomena” in chemistry—traditional chemistry knows of the existence of l but does not know that l precisely determines the intensity of regime switching.

Level 2: Observable regime factors—period and valence_electrons. period and valence_electrons are two orthogonal projections of l onto observable dimensions. They were automatically selected by exhaustive testing directly from the data—the program, knowing no chemical concepts, automatically locked them as regime factors from among 27 candidate factors. Traditional chemistry found these two dimensions through empirical induction but did not know they are projections of l —traditional chemistry knows the “what”; the Factor Hierarchy Law reveals the “why.”

Level 3: Super-probe—electronegativity. Electronegativity is the only observable that can simultaneously mark the same regime-switching center across three regime axes (s/p, d, f). Tang (2026s) discovered that the triple breaks of electronegativity on the number axis ($Z=54, 72, 86$) precisely mark the complete boundary of the f-block regime switch. Electronegativity has the deepest coupling with regime switching—it is the “optimal projection” of l in the space of chemical properties.

This three-level hierarchical structure constitutes the core theoretical framework of the Factor Hierarchy Law in chemistry and is the starting point for the derivation of the Chemical Gauge Group Theorem in this paper.

4 The Chemical Gauge Group Theorem

4.1 Statement of the Theorem

Chemical Gauge Group Theorem (Theorem 2, MA3-2): Let the meta-regime factor valence_electrons possess $n=8$ mutually exclusive regime states (outermost electron numbers = 1, 2, ..., 8), and let its moderating effects cover all executive factors with irreducible interaction effects. Then the regime-state structure of valence_electrons, under the symmetry constraint of the three-dimensional rotation group $SO(3)$, necessarily produces, and only produces, the four irreducible representations $l=0, 1, 2, 3$ —namely, the s, p, d, and f orbitals. No regime state with $l \geq 4$ exists.

4.2 Complete Derivation

Table 1 reports the seven-step derivation process of the Chemical Gauge Group Theorem, with each step annotated with its premise source and derivation logic.

Table 1: Seven-Step Derivation of the Chemical Gauge Group Theorem (MA3-13)

Step	Content	Premise Source	Derivation Logic
1	valence_electrons is a meta-regime factor with 8 mutually exclusive regime states	Tang (2026s) Axiom 1 verification: exhaustive testing automatic selection + interaction effect matrix 14/27 significant	Empirical confirmation
2	According to the Gauge Group Structure Theorem (Tang, 2026p, Theorem 4), any meta-regime factor with n mutually exclusive regime states necessarily produces an $SU(n)$ gauge group	Mathematical proof of Theorem 4 (based on the Lie group classification theorem)	Mathematical derivation from axiom to theorem
3	However, chemical systems differ fundamentally from particle physics systems: the underlying symmetry of electron configuration is not abstract unitary transformations but the irreducible representations of the three-dimensional rotation group $SO(3)$	Standard conclusion of quantum mechanics: electrons move in a spherically symmetric Coulomb potential, the angular part of the wavefunction is the spherical harmonics $Y_{l,m}(\theta, \phi)$, and the symmetry group is $SO(3)$	Adoption of physical fact
4	The unique realization of $SU(8)$ under the $SO(3)$ symmetry constraint is to decompose it into a direct sum of irreducible representations of $SO(3)$	Standard conclusion of group representation theory: any finite-dimensional representation of a compact Lie group can be decomposed into a direct sum of irreducible representations	Group-theoretic mathematical derivation
5	The irreducible representations of $SO(3)$ are labeled by the orbital angular momentum quantum number l : $l=0$ (1-dim), $l=1$ (3-dim), $l=2$ (5-dim), $l=3$ (7-dim), $l=4$ (9-dim)...	Standard conclusion of $SO(3)$ representation theory: the dimension of the l -th irreducible representation is $2l+1$	Group-theoretic mathematical derivation
6	The sum of the dimensions of the irreducible representations must exactly accommodate 8 mutually exclusive regime states. The sum of dimensions for $l=0,1,2,3$ is $1+3+5+7=16$, which, after accounting for double spin degeneracy, exactly corresponds to 8 regime states	Constraint of 8 regime states from Step 1 + dimension formula from Step 5	Combinatorial constraint solution
7	No regime state with $l \geq 4$ exists. The $l=4$ representation is 9-dimensional, corresponding to 18 spin-orbitals, far exceeding the capacity of 8 regime states. To accommodate $l=4$, at least 9 mutually exclusive regime states would be required—this	Dimensional constraint from Step 6 + nuclear stability quantitative criterion from Tang (2026t)	Physical limit constraint

Step	Content	Premise Source	Derivation Logic
	would necessitate a nuclear charge number Z exceeding the nuclear stability limit ($Z \gg 126$, fission barrier height dropping below 1 MeV)		

4.3 Group-Theoretic Correspondence between $SO(3)$ Irreducible Representations and Regime States

The correspondence between the irreducible representations of $SO(3)$ and chemical orbitals is shown in Table 2.

Table 2: Correspondence between $SO(3)$ Irreducible Representations and Chemical Orbitals (MA3-10, MA3-11)

l	Irreducible Representation Dimension ($2l+1$)	Orbital Name	Number of Regime States after Spin Degeneracy	Corresponding Element Block
0	1	s	2	s-block
1	3	p	6	p-block
2	5	d	10	d-block
3	7	f	14	f-block
4	9	g	18 (predicted, but exceeds the capacity of 8 regime states)	Does not exist

The sum of the dimensions of the irreducible representations is $1+3+5+7=16$. After accounting for double spin degeneracy, the 16 spin-orbitals exactly correspond to 8 spin-degenerate regime states. This correspondence is not a coincidence—it is the unique possible group-theoretic realization of the 8 regime states of valence_electrons under the $SO(3)$ symmetry constraint.

4.4 Isomorphism with and Difference from Theorem 4 (Gauge Group Structure Theorem)

The Chemical Gauge Group Theorem is completely isomorphic in logical structure to the Gauge Group Structure Theorem (Theorem 4) proved by Tang (2026p) in particle physics, but there is a fundamental difference.

In particle physics, the derivation path of Theorem 4 is: color charge is a meta-regime factor with a three-state structure $\rightarrow$ Lie group classification theorem $\rightarrow$ $SU(3)$ is the unique gauge group satisfying the conditions. This derivation is purely mathematical—it requires only the Lie group classification theorem, an established mathematical fact, and no additional physical assumptions.

In chemistry, the derivation path of the Chemical Gauge Group Theorem is: valence_electrons is a meta-regime factor with 8 regime states $\rightarrow$ irreducible representations of $SO(3)$ are needed to accommodate

these regime states $\rightarrow$ SO(3) irreducible representations are labeled by $l \rightarrow l=0,1,2,3$ exactly provide 16 spin-orbitals $\rightarrow$ s, p, d, f orbitals are the unique possible solution.

These two derivation paths share the same starting point (the regime-state structure of the meta-regime factor) and the same intermediate step (representation theory of Lie groups/compact Lie groups), but the chemical derivation requires the additional introduction of a physical premise: the underlying symmetry of electron configuration is SO(3). This premise is not derived by the Factor Hierarchy Law itself—it is an already-established standard conclusion of quantum mechanics.

Therefore, the Chemical Gauge Group Theorem is positioned as a “synthetic derivation”—the Factor Hierarchy Law provides the regime-state structure, quantum mechanics provides the SO(3) symmetry constraint, and the two together derive the inevitability of s, p, d, and f orbitals. This distinction has been candidly noted in both the naming of the theorem and the main text.

4.5 Corollary: The Impossibility of g Orbitals

A direct corollary of the Chemical Gauge Group Theorem is that g orbitals ($l=4$) cannot appear within the framework of the existing periodic table.

From a group-theoretic perspective, the $l=4$ irreducible representation of SO(3) is 9-dimensional, corresponding to 18 spin-orbitals, requiring at least 9 mutually exclusive regime states to accommodate. However, valence_electrons has only 8 mutually exclusive regime states (outermost electron numbers from 1 to 8)—18 spin-orbitals far exceed the capacity of 8 regime states.

From a physical perspective, for g orbitals to appear, valence_electrons would need to possess at least 9 mutually exclusive regime states. However, this would require a nuclear charge number Z exceeding the nuclear stability limit—according to calculations based on the liquid drop model and shell corrections, for nuclides with $Z>126$, the fission barrier height rapidly drops below 1 MeV, making it impossible to form measurable chemical properties (Tang, 2026t). The end of the periodic table is not an arbitrary human stipulation but the inevitable joint consequence of topological constraints from regime centers and the physical limits of nuclear stability—this is the core content of Theorem 4 (MA3-4).

It must be candidly stated that the above argument rigorously proves mathematically that “under the SO(3) symmetry constraint, 8 regime states can only correspond to the four irreducible representations $l=0,1,2,3$.” It does not prove that “SO(3) symmetry is the only possible symmetry”—if there exist new symmetry structures beyond standard quantum mechanics, the 8 regime states might correspond to different types of orbitals. The Chemical Gauge Group Theorem is rigorous within the currently known physical framework, but its universality is limited by the scope of applicability of SO(3) symmetry.

5 Division of Labor and Subsumption between the Factor Hierarchy Law and Quantum Mechanics

5.1 What Quantum Mechanics Can Explain

Quantum mechanics provides the “executive-layer” rules for the periodic table. The Schrödinger equation yields the spatial distribution of electron wavefunctions (spherical harmonics). The Pauli exclusion principle stipulates that each orbital can accommodate at most two electrons. Hund’s rules dictate the order in which electrons fill orbitals. Together, these rules determine the basic pattern of electron configuration—which orbitals fill first, which fill later, and how many electrons each orbital can hold.

Quantum mechanics can explain: why the first period has only 2 elements (the 1s orbital can hold only 2 electrons), why the second period has 8 elements (2s+2p, 4 orbitals, holding 8 electrons), and why the third period also has 8 elements (3s+3p, 4 orbitals, holding 8 electrons, with the 3d orbital unfilled due to higher energy).

Table 3 summarizes the boundary of division of labor between quantum mechanics and the Factor Hierarchy Law in the field of chemistry.

Table 3: Boundary of Division of Labor between the Factor Hierarchy Law and Quantum Mechanics (MA3-14)

Question	Can Quantum Mechanics Answer?	Can the Factor Hierarchy Law Answer?	Division of Labor
How do electrons fill orbitals?	✓ Yes (Schrödinger equation + Pauli principle + Hund’s rules)	—	Exclusive to quantum mechanics
Why does a regime switch occur at $Z=20$?	✗ No (only knows that 3d orbitals start filling from there)	✓ Yes (singularity center, Chow $F=1.53 \times 10^{23}$)	Exclusive to the Factor Hierarchy Law
The specific intensity of the switch at $Z=20$?	✗ No	✓ Yes ($F=1.53 \times 10^{23}$, precisely quantified)	Exclusive to the Factor Hierarchy Law
The f-block/d-block switching intensity ratio?	✗ No	✓ Yes ($F_f/F_d=2.20 \approx (3/2)^2$)	Exclusive to the Factor Hierarchy Law
The intrinsic transition bandwidth?	✗ No	✓ Yes ($\Delta P=3.0$, four properties completely consistent)	Exclusive to the Factor Hierarchy Law
Why does the periodic table have an end?	✗ No (only says nuclei are unstable)	✓ Yes (topological constraint of regimes + nuclear stability limit)	Exclusive to the Factor Hierarchy Law
Why exactly s, p, d, f orbitals?	✗ No (only says $l=0,1,2,3$ have solutions, but cannot explain why no $l \geq 4$)	✓ Yes (Chemical Gauge Group Theorem)	Exclusive to the Factor Hierarchy Law

5.2 What Quantum Mechanics Cannot Explain

Quantum mechanics cannot explain the laws of the “regime layer”—it cannot tell you why regime switches of the 10^{23} magnitude occur at specific positions ($Z=20$ and $Z=57$), it cannot tell you why the f-block switching intensity is 2.20 times that of the d-block, it cannot tell you why the transition bandwidth is uniformly 3.0 periods, and it cannot tell you why the periodic table has an end.

More fundamentally: quantum mechanics cannot explain why there are exactly four types of atomic orbitals—s, p, d, and f. From the spherical harmonics $Y_{\{l,m\}}(\theta,\phi)$ solved from the Schrödinger equation, l can take any non-negative integer— $l=0, 1, 2, 3, 4, \dots$ with no mathematical upper bound. Quantum mechanics can only say “we have only observed $l=0, 1, 2, 3$ ” but cannot explain “why there is no $l \geq 4$.” The traditional answer is “because the 118 known elements do not require g orbitals to fill electrons”—but this is a circular argument; it does not explain why the number of elements happens to terminate precisely at the range that does not require g orbitals.

5.3 Regime Switching as the Inevitable Emergence of Quantum Mechanics on Finite Systems

The relationship between the Factor Hierarchy Law and quantum mechanics is not one of competition but of division of labor and subsumption. Quantum mechanics provides the “executive-layer” rules—it tells you how electrons are configured within a given orbital structure. The Factor Hierarchy Law provides the “regime-layer” rules—it tells you how the orbital structure itself is determined by regime switching.

More precisely: quantum mechanics is a complete theory in the $Z \rightarrow \infty$ limit—in this limit, all possible orbital types ($l=0, 1, 2, 3, 4, \dots$) are mathematically allowed. However, in the finite- Z range, the topological constraints of regime switching limit the actually realizable orbital types—g orbitals ($l=4$) would require more than 8 mutually exclusive regime states, which would require a nuclear charge number Z exceeding the nuclear stability limit.

Therefore, regime switching is not new physics beyond quantum mechanics—it is the inevitable emergence of quantum mechanics on finite systems. Quantum mechanics provides the mathematical framework of “infinite possibilities”; the Factor Hierarchy Law reveals the physical constraints of “finite realizations.”

This relationship is structurally completely isomorphic to the argument of Tang (2026m) in statistical physics: traditional universality is a mathematical convergence product under the thermodynamic limit ($L \rightarrow \infty$), while regime boundaries are the fundamental quantities in finite systems. In chemistry, the inevitability of s, p, d, f is a mathematical abstraction in the $Z \rightarrow \infty$ limit (all l are mathematically allowed), and regime switching explains why, in the finite- Z range, “only $l=0,1,2,3$ ” can be physically realized.

This argument constitutes Theorem 5 (MA3-5): **The Factor Hierarchy Law subsumes quantum mechanics—demoting quantum mechanics from a “fundamental theory” to “a provider of symmetry constraints as physical input.”**

6 The Demotion of Gauge Field Theory

6.1 Gauge Field Theory as the Expression of Regime Boundaries in the Continuous

Symmetry Limit

Tang (2026q) argued in particle physics that gauge field theory is not the most fundamental theoretical framework—it is the expression of regime boundaries in group-theoretic language. The Factor Hierarchy Law is a deeper organizing principle than gauge field theory. The core logic of this argument is: gauge field theory cannot derive the Factor Hierarchy Law (it cannot explain why the Mass coefficient jumps by a factor of 19 at the charm quark threshold), but the Factor Hierarchy Law can derive gauge field theory (three-state structure of the meta-regime factor $\rightarrow$ SU(3) gauge group).

In chemistry, the identical logical structure applies perfectly. Quantum mechanics cannot derive the Factor Hierarchy Law—it cannot explain why the density–atomic mass correlation plummets from 0.73 to 0.28 at $Z=20$, or why the transition bandwidth is uniformly 3.0 periods. However, the Factor Hierarchy Law can derive the core conclusions of quantum mechanics related to the periodic table—the 8 regime states of valence_electrons + SO(3) symmetry constraint $\rightarrow$ the inevitability of s, p, d, f orbitals.

Therefore, the corresponding status of quantum mechanics in chemistry is demoted to “a provider of symmetry constraints as physical input.” It is not overthrown—it is demoted. The mathematical structure of quantum mechanics (Schrödinger equation, spherical harmonics, Pauli principle) remains correct, but it is no longer “the most fundamental.” The Factor Hierarchy Law is the deeper organizing principle, and quantum mechanics is its specific expression in the continuous symmetry limit.

6.2 The Factor Hierarchy Law as the Deeper Organizing Principle

The core of this demotion argument is the argument about causal direction. Quantum mechanics can describe the electron configuration at each Z but cannot explain why regime switches occur at specific Z values. The Factor Hierarchy Law, through the Chow test, not only detects these regime switches but also precisely quantifies their intensities and positions.

More fundamentally: quantum mechanics cannot derive the inevitability of s, p, d, f from first principles—it can only say “we have observed $l=0,1,2,3$.” The Chemical Gauge Group Theorem, starting from the axioms of the Factor Hierarchy Law, rigorously proves that under the SO(3) symmetry constraint,

the 8 regime states can only correspond to these four irreducible representations. The causal direction is reversed—it is not because quantum mechanics solved s, p, d, f that the periodic table is so arranged, but because the regime-state structure of valence_electrons determines that there must be, and only be, these four types of orbitals, with quantum mechanics merely providing the mathematical tool for realizing this structure.

6.3 Complete Isomorphism with the Argument of Tang (2026q)

Table 4 reports the isomorphic comparison between the demotion of gauge field theory and the reduction of universality.

Table 4: Isomorphic Comparison between the Demotion of Gauge Field Theory and the Reduction of Universality (MA3-15)

Dimension	Statistical Physics (Tang, 2026m)	Particle Physics (Tang, 2026q)	Chemistry (This Paper)
Demoted traditional theory	Universality (critical exponents independent of microscopic details)	Gauge field theory (SU(3)×SU(2)×U(1))	Quantum mechanics (Schrödinger equation + Pauli principle)
Effective range of traditional theory	$L \rightarrow \infty$ thermodynamic limit	Continuous field theory limit	$Z \rightarrow \infty$ limit
Fundamental quantities in finite systems	Regime boundaries (double-peak fission, precursor boundaries)	Regime boundaries (charm, bottom, strange regions)	Regime switching (d-block, f-block singularity centers)
Status after demotion	Universality demoted to a statistical approximation at the limit	Gauge field theory demoted to the expression of regime boundaries in group theory	Quantum mechanics demoted to a provider of symmetry constraints as physical input
Role of the Factor Hierarchy Law	True fundamental quantities in finite systems	Deeper organizing principle than gauge field theory	Deeper organizing principle than quantum mechanics

The isomorphic arguments across three disciplines jointly point to one conclusion: the “fundamental concepts” of traditional theories are mathematical products of idealized limits, whereas the Factor Hierarchy Law describes the true structures in finite systems. This is yet another piece of evidence for the Factor Hierarchy Law as a cross-disciplinary universal law.

7 The Chemical Higgs Mechanism

7.1 The Function of the Higgs Mechanism in Particle Physics

In the Standard Model of particle physics, the core function of the Higgs mechanism is: through the spontaneous symmetry breaking of a scalar field (the Higgs field), to endow elementary particles with mass. The W and Z bosons acquire mass through coupling with the Higgs field, and fermions acquire mass through Yukawa coupling. The Higgs mechanism partitions a continuous mass spectrum into discrete mass

eigenstates—the electron, muon, tau, up quark, down quark, charm quark, strange quark, top quark, and bottom quark each have their specific mass values.

7.2 The Isomorphic Function of Regime Switching in Chemistry

In chemistry, regime switching plays a completely isomorphic function. When the orbital angular momentum l crosses critical values (from $l=1$ to $l=2$, i.e., from the p-block to the d-block; from $l=2$ to $l=3$, i.e., from the d-block to the f-block), the electron configuration rules undergo a regime switch. This switch partitions the continuous nuclear charge spectrum Z into discrete property matrices—the s-block, p-block, d-block, and f-block each have their specific chemical property patterns.

The Higgs mechanism endows particles with the attribute of “mass”; regime switching endows elements with the attribute of “chemical properties.” Both are mechanisms of symmetry breaking that transform continuous parameters into discrete structures. The Higgs mechanism generates a mass spectrum; regime switching generates a property spectrum. The Higgs mechanism produces resonance peaks at critical points (experimentally detectable singularity signals); regime switching produces singularity peaks in the Chow F at critical points (statistically detectable signals at the 10^{23} magnitude).

This is not a rhetorical analogy but a mathematical isomorphism. Both involve: a continuous parameter (the vacuum expectation value of the Higgs field / the nuclear charge number Z) $\rightarrow$ a symmetry-breaking mechanism ($SU(2) \times U(1) \rightarrow U(1)$ / $SO(3)$ irreducible representations) $\rightarrow$ discrete physical outputs (mass eigenstates / orbital types).

This isomorphism constitutes Theorem 6 (MA3-6): **Regime switching is the chemical version of the Higgs mechanism—it endows elements with regime identities at specific critical points, generates regime boundaries, and thereby partitions the continuous nuclear charge spectrum into discontinuous property matrices.**

7.3 Endowing Elements with Regime Identities—Partitioning the Continuous Spectrum into Discrete Matrices

Under the framework of the chemical Higgs mechanism, the properties of each element are not continuously determined by its nuclear charge number Z but are determined by the regime to which it belongs. Within the same regime, properties change smoothly and predictably with Z ; at regime boundaries, the property-determination equation systematically reset—the coefficients of the old equation no longer apply, and the coefficients of the new equation begin to take effect.

This is why 95 elements (80%) are in the “hyper-unstable” state—they happen to be situated on regime boundaries or at regime nexus points, being simultaneously pulled by multiple sets of rules. This is also why the traditional periodic law requires 16 “exceptions” to patch—because these exceptional elements happen

to be situated on regime boundaries, and their properties cannot be described by the rules of a single regime. The chemical Higgs mechanism reinterprets these “exceptions” as inevitable products of regime switching—exceptions no longer exist; only statistical inevitabilities at regime boundaries exist.

8 The End of the Periodic Table

8.1 The Chemical Version of the Three-Generation Proposition—Topological

Constraints of Transition Bandwidths

Tang (2026p) proposed the Three-Generation Proposition (Proposition 6) in particle physics: the topological constraints of transition bandwidths and the experimental exclusion by the LHC jointly point to the number of heavy quarks being limited to four. In chemistry, a completely isomorphic logic applies: the topological constraints of transition bandwidths determine the upper limit of the number of regime centers.

Table 5 reports the quantitative relationship between the spacing of regime centers and transition bandwidths.

Table 5: Progressive Expansion of Regime Center Spacing and Transition Bandwidths

Regime Switch	Change in l	Entrance Z	Spacing (number of elements)	Internal Transition Bandwidth (periods)	Singularity Intensity Ratio
$s \rightarrow p$	$0 \rightarrow 1$	$Z=3$	2 (H to B)	≈ 0	—
$p \rightarrow d$	$1 \rightarrow 2$	$Z=21$	11 (Ne to Sc)	≈ 2	F_d/F_p (not measured)
$d \rightarrow f$	$2 \rightarrow 3$	$Z=57$	37 (Ca to La)	3.0	$F_f/F_d=2.20 \approx (3/2)^2$
$f \rightarrow g$ (predicted)	$3 \rightarrow 4$	$Z \approx 121$	64 (La to Ubu)	3.0 (predicted)	$F_g/F_f=(4/3)^2 \approx 1.78$
$g \rightarrow h$ (impossible)	$4 \rightarrow 5$	$Z \gg 126$	Far exceeds nuclear stability limit	Far exceeds nuclear stability limit	—

The spacing expands exponentially: $2 \rightarrow 11 \rightarrow 37 \rightarrow 64 \rightarrow$ far beyond the nuclear stability limit. This progressive expansion is not accidental—it is an inevitable consequence of the dramatic increase in the complexity of orbital spatial forms as the orbital angular momentum l increases. The larger l is, the more nodes the orbital has, the more complex its spatial form, and the more elements are needed for the regime switch to complete the transition.

8.2 The g-Block as the Last Detectable Regime Center

The entrance to the g-block ($l=4$) at $Z \approx 121$ is the last possibly detectable regime center. Tang (2026t) provided a quantitative prediction of the g-block entrance Chow $F \approx 5.99 \times 10^{23}$ and the eighth-period transition bandwidth of 3.0 periods.

However, the existence of the g-block itself is also constrained by the nuclear stability limit. According to calculations based on the liquid drop model and shell corrections, $Z\approx 121\text{--}126$ may fall within the “island of stability”—additional binding energy provided by nuclear shell effects. However, for nuclides with $Z>126$, the fission barrier height rapidly drops below 1 MeV, below the competitive thresholds of alpha decay and spontaneous fission. This means: the first half of the g-block elements ($Z\approx 121\text{--}126$) may have sufficiently long half-lives for their chemical properties to be measured; the second half of the g-block elements ($Z\approx 127\text{--}138$) may have extremely short half-lives, making their chemical properties difficult to measure.

Therefore, the g-block is a “partially detectable” regime center—its first half may be verified in future superheavy element synthesis experiments, while its second half may never be fully observable.

8.3 Complete Isomorphism with the Argument of Tang (2026p)

The argument for the end of the periodic table is completely isomorphic in logical structure to the Three-Generation Proposition of Tang (2026p). Both rely on: topological constraints of transition bandwidths (constraining the number of heavy quarks in particle physics, constraining the number of orbital types in chemistry) + physical limit constraints (LHC exclusion of fourth-generation quarks in particle physics, nuclear stability limits on superheavy elements in chemistry).

This argument constitutes Theorem 4 (MA3-4): **The end of the periodic table—the g-block is the last detectable regime center, and the h-block can never exist due to the nuclear stability limit.**

9 The Five-Layer Axiomatic Structure

9.1 Overview of the Five-Layer Structure

Based on all the empirical discoveries of Tang (2026s), the systematic construction of Tang (2026t), and all the theoretical derivations of this paper, the regime-switching theory of the periodic law presents a rigorous five-layer axiomatic structure. Table 6 reports the complete mapping of this structure.

Table 6: Complete Mapping of the Five-Layer Axiomatic Structure in Chemistry (MA3-7)

Layer	Name	Core Content	Specific Embodiment in Chemistry
Layer 1	Law Layer (Axiom System)	Three axioms	Axiom 1: period and valence as regime factors; Axiom 2: $Z=20$ and $Z=57$ as singularity centers; Axiom 3: 14/27 interaction effects significant
Layer 2	Norm Layer (Operational Standards)	Six functional dimensions	Search completeness (exhaustive 27 factors), moderation relationship (interaction matrix), structural breaks (10 tools), causal direction (conditional independence + DAG), problem completeness (5W2H), process closure (PDCA)
Layer 3	Algorithm Layer (Operational)	Three-step MRSD logic	Input 118-element objective data → structural optimization (10-tool exhaustive search) → output regime annotation

Layer	Name	Core Content	Specific Embodiment in Chemistry
	Implementation)		table
Layer 4	Theorem and Proposition Layer (Mathematical Proofs)	Four quantitative laws + Chemical Gauge Group Theorem	Calcium-Lanthanum Singularity Theorem ($F(l) \propto l^2$), Topological Uniformity of Intrinsic Transition Bandwidths ($\Delta P=3.0$), Translational Invariance of valence ($CV=0$), Chemical Gauge Group Theorem ($SO(3) \rightarrow s,p,d,f$)
Layer 5	Theory Layer (Physical Interpretation)	Nine cross-disciplinary corollaries	See §10.2

9.2 Comparison with the Five-Layer Structure of Particle Physics

Tang (2026q) established a completely isomorphic five-layer axiomatic structure in particle physics: Law Layer (three axioms) → Norm Layer (six functional dimensions) → Algorithm Layer (three-step MRSD logic) → Theorem and Proposition Layer (five theorems + one proposition) → Theory Layer (eight corollaries).

The comparison between the chemical five-layer structure and the particle physics five-layer structure reveals that the Factor Hierarchy Law unfolds across different disciplines following the same logical structure. The three axioms of the Law Layer are cross-disciplinarily invariant—they describe the universal logic of the functional stratification of variables in any complex system based on objective data. The six functional dimensions of the Norm Layer are cross-disciplinarily invariant—they are the standard framework for translating axioms into operational procedures. The three-step MRSD logic of the Algorithm Layer is cross-disciplinarily invariant—input objective data, perform structural optimization, output regime matrices. The Theorem and Proposition Layer is discipline-specific—it derives, from the axioms and in combination with the physical facts of the specific discipline, the core theorems of that discipline. The Theory Layer is discipline-specific—it integrates the theorems into a self-consistent physical picture.

10 The Final Argument for Cross-Disciplinary Unification

10.1 Cross-Disciplinary Isomorphic Evidence Matrix across Five Disciplines

Table 7 reports the isomorphic evidence matrix of the Factor Hierarchy Law across five completely independent disciplines.

Table 7: Isomorphic Evidence Matrix across Five Disciplines (MA3-8)

Dimension	Finance	Astronomy	Statistical Physics	Particle Physics	Chemistry
Regime Factor	China-U.S. yield spread	Stellar spectral type	Temperature, disorder	Color charge	period, valence_electrons
Executive Factors	VIX, DXY	Transit depth	M_sp, U ₄ , χ	Mass, Spin, Isospin	Density, electronegativity, ionization energy
Regime	Rule-reset	Rule-reset	Rule-reset +	Rule-reset +	Direction-reversal +

Switching Type			direction-reversal	direction-reversal	mixed
Singularity Center	Spread $\approx$ 0	Tang Break (4762 K)	Tc (2D/3D Ising)	Charm region (2.29 GeV), Bottom region (9.46 GeV)	Z=20 (d-block), Z=57 (f-block)
Transition Bandwidth	—	—	0.1 (3D Ising L=64)	0.085–3.517 GeV	3.0 periods
Precursor Cascade	—	—	M _{sp} dimension non-convergent	Strange T ₂ $\approx$ Charm T ₁ , Charm T ₂ =Bottom precursor	Ionization energy period=2.0 precursor
Gauge Group Structure	Institutional interface	—	—	SU(3) (Theorem 4)	SO(3) irreducible representations
Testable Prediction	Anchor n=3	—	—	Three-Generation Proposition (Proposition 6)	g-block F $\approx$ 5.99 $\times$ 10 ²³

The “—” entries in the table indicate that the discipline has not yet completed verification on that dimension. This does not affect the cross-disciplinary universality of the conclusions—it merely indicates directions for future research.

10.2 The Factor Hierarchy Law as the “Law of Meta-Laws”

Tang (2026r) positioned the Factor Hierarchy Law as the “law of meta-laws”—it reveals the universal logic of the functional stratification of variables in any complex system based on objective data, its scope far exceeding that of deduction-type meta-laws such as Noether’s theorem, the principle of least action, and Shannon’s information theory, because it requires no theoretical presuppositions and depends only on the objectivity of the data.

Chemistry is the fifth discipline conquered by the Factor Hierarchy Law. The isomorphic evidence matrix across five disciplines proves that this methodology is not a contingent product of finance, not a special tool of astronomy, not a mathematical coincidence of statistical physics, not domain-dependent on particle physics, and not an empirical induction of chemistry—it is the universal logic of the functional stratification of variables in any complex system based on objective data.

The nine cross-disciplinary corollaries (Theorem 9, MA3-9) constitute the complete theoretical system of the Factor Hierarchy Law in chemistry:

1. The periodic table is a three-regime system cut by dual singularity centers;
2. The traditional periodic law is the expression of regime boundaries in the language of groups and periods, and is not the most fundamental law;
3. The Factor Hierarchy Law is a deeper organizing principle than the periodic law;
4. The orbital angular momentum l is the meta-regime factor hidden behind period and valence;
5. Regime switching is the chemical version of the Higgs mechanism—generating regime boundaries and partitioning the continuous property spectrum;
6. The three breaks of the first ionization energy are the chemical version of the precursor cascade;

7. The inv and cube transformations are the chemical version of activation operators;
8. The end of the periodic table is the joint consequence of topological constraints of regimes and the nuclear stability limit;
9. The irreducible predictive uncertainty at regime boundaries is the ultimate boundary of chemical theory.

11 Testability Program

Table 8 reports the four testable predictions of the Factor Hierarchy Law in chemistry.

Table 8: Testability Program (MA3-18)

Prediction	Core Content	Test Pathway	Falsifiability Condition
Prediction 1	The inv-transformed Chow F at the g-block entrance ($Z \approx 121-138$) is approximately 5.99×10^{23}	Apply the inv transformation to the properties of superheavy elements, scan for structural breaks on the number axis using SupF	F-value falls outside $3.0-9.0 \times 10^{23} \rightarrow F(I) \propto I^2$ is falsified
Prediction 2	The intrinsic property transition bandwidth of the eighth period is 3.0 periods	Perform Bai-Perron scans on the density, electronegativity, ionization energy, and atomic mass of eighth-period elements, and calculate breakpoint spans	The span of any property is <2.0 or >4.0 periods $\rightarrow$ topological cointegration is falsified
Prediction 3	The electronegativity of element 119 rebounds to 0.85–0.90, and element 120 exhibits pseudo-noble character ($I_1 \approx 5.8-6.1$ eV)	Experimentally synthesize elements 119/120 and measure their electronegativity and first ionization energy	Electronegativity of 119 is <0.7 or I_1 of 120 is <5 eV $\rightarrow$ URSO extrapolation is falsified
Prediction 4	The equally spaced regime grid on the valence_electrons axis continues in superheavy elements	Scan for significant breakpoints on the valence_electrons axis using 6 tools on superheavy element data	Non-integer breakpoints appear or spacing $\neq 1 \rightarrow$ translational invariance is falsified

12 Historical Positioning and Conclusion

12.1 The Triadic Historical Positioning

The development of the periodic table has passed through three eras (Theorem 9, MA3-9):

Mendeleev (1869) discovered the “form” of the periodic law. Through empirical induction, he arranged the 63 elements then known by atomic weight and discovered the periodic variation of chemical properties. He boldly left vacant spaces for unknown elements and predicted their properties—one of the greatest empirical inductions in the history of science. However, Mendeleev did not know why the periodic law holds, nor did he know why there are exceptions.

Moseley (1913) found the “substance” of the periodic law. Using X-ray spectroscopy, he established that the atomic number Z is the true basis of periodicity, correcting Mendeleev’s error of using atomic

weight. The periodic law was revised from “a periodic function of atomic weight” to “a periodic function of atomic number.” However, Moseley still did not know why the periodic law holds; he only knew what its basis is.

This theory (2026) has discerned the “spirit” of the periodic law. It proves that the periodic law is not the most fundamental law but the projection of regime switching in the language of groups and periods. The regime factors (period and valence_electrons) are two orthogonal projections of the meta-regime factor (orbital angular momentum l) onto observable dimensions. When l crosses critical values, regime switching occurs, and the properties of elements on the periodic table are systematically reset. The traditional periodic law is demoted from a fundamental law to a statistical approximation in the $Z \rightarrow \infty$ limit, while regime switching becomes the true fundamental quantity in the finite- Z range.

12.2 Conclusion

Building on the empirical discoveries of the two preceding papers, this paper has completed three core theoretical derivations. First derivation: the Chemical Gauge Group Theorem—starting from the eight mutually exclusive regime states of valence_electrons, through the irreducible representations of $SO(3)$, rigorously proving that s, p, d, and f orbitals are the unique possible solution. Second derivation: the division of labor and subsumption between the Factor Hierarchy Law and quantum mechanics—the Factor Hierarchy Law provides the regime-state structure, quantum mechanics provides the $SO(3)$ symmetry constraint, and the two together derive the inevitable emergence of the periodic law. Third derivation: the end of the periodic table—the g-block is the last detectable regime center, and the h-block can never exist due to the nuclear stability limit.

Based on these three derivations, this paper has established the five-layer axiomatic structure of the regime-switching theory of the periodic law—Law Layer $\rightarrow$ Norm Layer $\rightarrow$ Algorithm Layer $\rightarrow$ Theorem and Proposition Layer $\rightarrow$ Theory Layer. This structure is completely isomorphic to the five-layer structure established by Tang (2026q) in particle physics. The isomorphic evidence matrix across five disciplines proves: the Factor Hierarchy Law is not a contingent product of finance, not a special tool of astronomy, not a mathematical coincidence of statistical physics, not domain-dependent on particle physics, and not an empirical induction of chemistry—it is the universal logic of the functional stratification of variables in any complex system based on objective data.

Data Availability Statement

This paper is a theoretical derivation and system integration paper. All empirical foundations of the axioms and premises have been reported in detail in Tang (2026s) and Tang (2026t). All data are completely public, and any researcher can independently reproduce all empirical results.

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